AI-Powered Hospital Readmission Risk Prediction System

AI-driven clinical decision support that identifies patients at risk of 30-day readmission - fast, explainable, and ready for hospital pilots. Prepared for hospital administrators, clinical operations leaders, and healthcare data teams.

PATIENT SAFETY | COST REDUCTION | EXPLAINABLE AI

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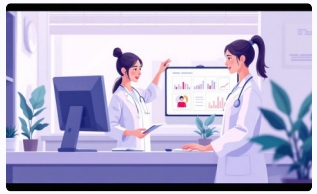
Problem & Solution - At a Glance

The Problem

11% of patients return within 30 days. Average cost per readmission ~ $15K - estimated $170M annual burden. Today: 0% proactively identified before discharge.



Our Solution

An AI-generated risk score (0–100%) + clear, actionable care plan delivered in 1 second to clinicians - enabling targeted interventions and estimated $100M+ savings at 60% recall.

Core question: Which patients will return within 30 days? → Our answer: an explainable AI risk score plus recommended actions at point of discharge.

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How It Works - Technical Pipeline

Input → Process → Output

01

Input

101K de-identified patient records, 50+ feature groups (demographics, prior admissions, vitals, meds, diagnoses).

02

Process

Data cleaning, imbalance handling, feature engine, XGBoost training, SHAP explainability for per-patient rationale.

03

Output

Risk score (0–100%), risk level (Low/Medium/High), top contributing features, and tailored recommendations - delivered in ~1 second.

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Model Performance - Comparison

Winner: XGBoost - highest recall, ideal where catching readmissions is priority. Precision remains modest; threshold tuning and workflow design required to balance workload.

XGBoost

Recall: 60.6% (best) • Precision: 17.8% • ROC-AUC: 0.671

Random Forest

Recall: 53.8% • Precision: 17.2% • ROC-AUC: 0.653

Logistic Regression

Recall: 54.4% • Precision: 16.3% • ROC-AUC: 0.638

Confusion matrix (optimized threshold): Predicted vs Actual - model captures ~60% of true readmissions (2,063 true positives) while producing false positives that must be managed through clinical triage.

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What Drives Readmission - Top Risk Factors

- number_inpatient - strongest predictor (history of prior admissions)
- discharge_disposition - discharge destination (home vs facility)
- total_visits - recent healthcare utilization
- payer_code_Unknown - insurance status uncertainty
- number_diagnoses - clinical complexity

Key insight: Prior inpatient stays and discharge destination are actionable - they guide discharge planning and post-acute referrals.

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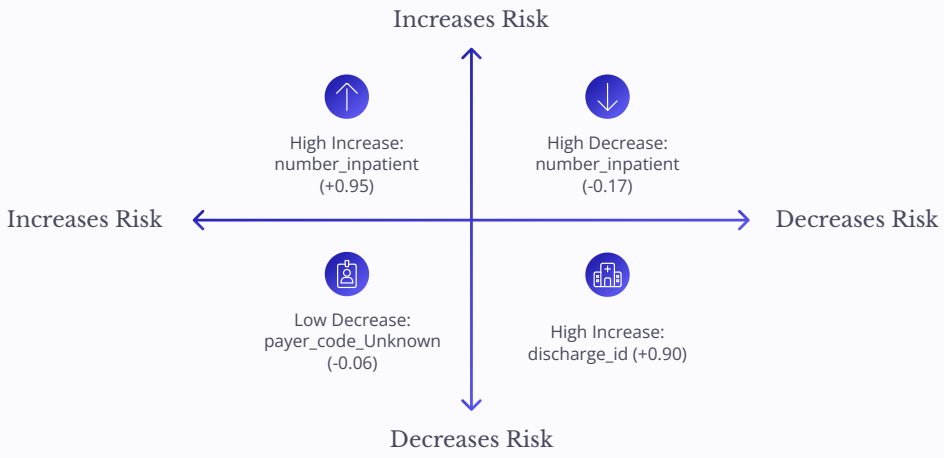
Explainable AI - SHAP Insights



SHAP reveals per-patient drivers:

- Increase risk: number_inpatient (+0.95), discharge_id (+0.90), total_visits (+0.85), payer_code_Unknown (+0.80)
- Decrease risk: number_inpatient (-0.17), payer_code_Unknown (-0.06), number_emergency (-0.01)

Explainability enables clinicians to see WHY the model flags a patient - building trust and guiding targeted interventions (e.g., arrange facility placement, intensified follow-up).

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Patient Case Study - From Data to Action

Patient snapshot: 1 prior inpatient stay, discharged to skilled facility, diabetic on insulin, 3 active medications.

Model: Risk score 53.4% → Medium risk.

Each recommendation is linked to the top SHAP contributors so clinicians understand the rationale.

1

1. Immediate

Schedule post-discharge follow-up within 7 days.

2

2. Medication

Perform comprehensive med reconciliation and insulin education.

3

3. Outreach

Weekly check-in calls for first 30 days; escalate if symptomatic.

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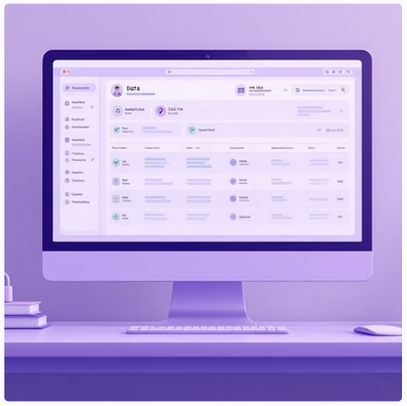
Validation -- Small-Scale Test Results

Five representative patient profiles were used to test alignment with clinical expectations. The model's risk progression matches clinical severity and utilization.

Patient Profile	Risk Score	Risk Level	Correct?
#1 Healthy (0 stays, 2 meds)	26%	LOW	✅
#2 Mild (1 stay, 12 meds)	54%	MEDIUM	✅
#3 Severe (3 stays, 25 meds)	68%	MEDIUM	✅
#4 Critical (5 stays, 35 meds)	73%	HIGH	✅
#5 Borderline (0 stays, 8 meds)	52%	MEDIUM	✅

Conclusion: Model outputs follow clinical logic and provide suitable triage signals for targeted review.



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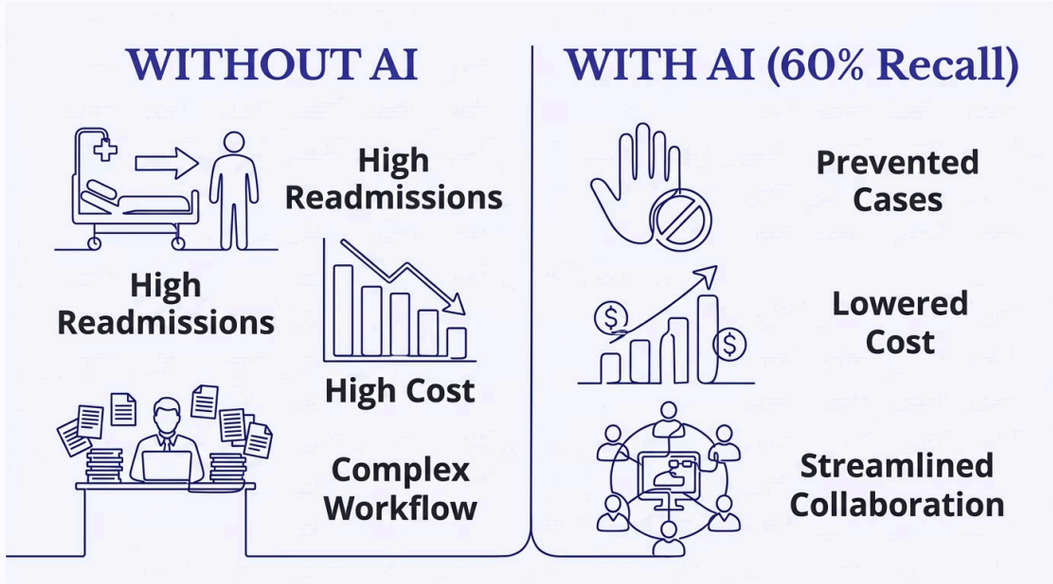
Real-World Impact - Clinical & Financial

Current baseline: ~11,357 readmissions annually (example hospital network), $170M attributable cost.

With AI (60% recall): ~6,882 readmissions prevented → estimated $103M saved. Clinician-facing tools include a Streamlit dashboard with downloadable patient reports.

- Primary users: Physicians, Case Managers, Nurses, Social Workers
- Workflow: Enter or auto-populate EHR data → 1-second risk + explanation → apply interventions



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Conclusion & Next Steps

What we delivered

- XGBoost model with 60.6% recall
- Interactive clinician dashboard
- SHAP-based explainability for each prediction
- Projected $100M+ potential savings at scale

Limitations & Recommended next steps

- Precision ~17.8% -require clinical triage workflow to manage false positives
- Pilot with 1 hospital to collect clinician feedback and outcome data
- Fine-tune threshold, retrain with new local data, and measure real-world impact
- Ensure clinician oversight, EHR integration, and privacy governance

Final verdict: 8/10 -- ready for a supervised hospital pilot. "AI doesn't replace doctors - it empowers them to catch more high-risk patients."

PILOT READY | CLINICAL OVERSIGHT

Thank you

D B S Samadhi • April 2026

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